MIL-M-38312B (USAF)
15 September 1988
SUPERSEDING
MIL-M-38312A (USAF)
4 August 1965
W/AMENDMENT I
15 December 1980

MILITARY SPECIFICATION

MANUALS, EQUIPMENT, REAL PROPERTY INSTALLED EQUIPMENT, PREPARATION OF

This specification is approved for use by the Ballistic Missile Office, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification establishes requirements for the organized assembly of maintenance and operations manuals, bulletins, brochures, and other data normally furnished by the manufacturer for his equipment, into manuals to be used for the operation and maintenance of components of real property installed equipment (RPIE).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Ballistic Missile Office/AWD, Norton AFB, CA 92409 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

ASMC F4530 AREA TMSS

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2. APPLICABLE DOCUMENTS

2.1 Government Documents.

2.1.1 Specifications and Standards. Unless otherwise specified, the following specifications and standards of the issue in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein (see 2.3.1).

SPECIFICATIONS

MILITARY

MIL-M-38784	Manuals, Technical: General Style and Format Requirements
MIL-P-38790	Printing Production of Technical Manuals General Requirement for (Photolithographic Negatives, Reproduction Assembly Sheets, Printing)

2.1.2 Other Government Documents, Drawings and Publications. The following other Government documents and publications form a part of this specification to the extent specified herein (see 2.3.2).

TECHNICAL ORDERS

TO 1-1-2	Corrosion Control and Treatment for Aerospace Equipment

2.2 Order of Precedence. In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence.

2.3 Source of Documents.

2.3.1 Government Specifications and Standards. Copies of the referenced federal and military specification and standards are available from the Department of Defense Single Stock Point, Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120. For specific acquisition functions, these documents should be obtained from the contracting activity or as directed by the contracting activity.

2.3.2 Other Government Documents. Copies of other government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity. The documents listed may be obtained as follows:

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a. Copies of listed documents are available from the Department of Defense Single Stock Point, Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

b. Copies of the GPO Style Manual should be obtained from the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

c. Copies of the DOD Thesaurus of Engineering and Scientific Terms and the DDC Retrieval and Indexing Terminology may be purchased from the National Technical Information Service, Springfield, VA 22161, or (for DOD activities and DOD contractors) from the Reference Services Branch, Defense Technical Information Center, Cameron Station, Alexandria, VA 22304-6145.

3. REQUIREMENTS

3.1 System Manual. RPIE system manuals shall be prepared for each RPIE functional system or homogeneous grouping of functional systems as identified by the contractor and approved by the contracting activity. The manual(s) shall provide instructions for operation and maintenance of the complete system. The functional breakout and the number of manuals required shall be determined by the contracting activity. (See 6.1)

3.1.1 Title Page. The title page shall be prepared under a relaxed format. The most economical, best available method shall be used (see Figure 1).

3.1.1.1 Identifying Number. The identifying number shall be positioned in the upper right hand corner. This number will be provided by the procuring agency.

3.1.1.2 Title. The title shall designate the type of manual, identify project and location applicability, and specify the facility, system or item of equipment covered in the manual.

3.1.1.3 Authority Notice. The title page shall contain an authority notice that will identify the agency having management responsibility for the operational system.

3.1.1.4 Contract Number. The contract number shall be shown to identify the contract under which the manual was prepared.

3.1.1.5 Publication Date. The publication date shall be the date on which the last material to be included was received (copy freeze date). The date shall be shown in the lower right corner.

3.1.2 List of Effective Pages. Immediately following the title page, there shall be a "List of Effective Pages" or "A" page. This page shall be prepared in accordance with MIL-M-38784.

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SAC CEM 35R-1-692-1

EQUIPMENT MANUAL

REAL PROPERTY INSTALLED EQUIPMENT

MISSILE WEAPON SYSTEM

USAF MODEL

LGM30

VANDENBERG AIR FORCE BASE

SUPPORT FACILITIES AS&I BUILDING

PUBLISHED UNDER AUTHORITY OF THE COMMANDER IN CHIEF, SAC

15 JANUARY 1974

FIGURE 1. Typical Title Page

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3.1.3 Index. Each manual shall be indexed alphabetically by noun. The listing by noun shall be by equipment nomenclature and shall contain the identification number (model, serial, type, etc.), the name of the manufacturer and the Master Equipment List (MEL) sequence number. The index shall be included in the front matter of the first volume of each manual immediately following the descriptive introductory material. The index shall not be required in succeeding volumes of multiple volume manuals (see Figure 2).

3.1.4 Art. In the event it is necessary to prepare art for equipment manuals, it shall be prepared in accordance with MIL-M-38784, except as otherwise specified herein. Every effort will be made to keep illustration pages no larger than 8-1/2 x 11 inches. Foldouts are to be avoided wherever possible.

3.1.5 Illustrations and Diagrams. Parts lists shall be supported by a cross section drawing, or other drawing, identifying every part of an item. Drawings to support the text shall also include an external view of the item showing main features unless the item is of such a nature that the parts list supporting drawings are adequate for positive identification. Diagrams shall include, where applicable, schematics for electrical and fluid flow and must be keyed to, or sufficient for, use with the text.

3.1.6 Master Equipment List (MEL) Sequence Identification. The first page of each individual manufacturer's brochure, bulletin, or other data shall be identified with the MEL sequence number assigned to the item.

3.1.7 Item Identification. If an individual brochure provides information on more than one item, the information applicable to the installed item shall be indicated by an arrow. The arrow may be made by the most expedient and economical method so long as it is clearly legible.

3.1.8 Relaxed Format and Reproduction Requirements. Relaxed format and economical reproduction methods shall be permitted for any material that must be prepared. Areas which may be relaxed are as follows:

a. Use of single column format (type continuous across page).

b. Use of standard typewriter to prepare reproduction copy.

c. Use of free hand lettering on illustrations.

d. Use of office-type reproduction equipment.

e. Use of nonuniform lettering size on final copy. (Not smaller than 6-point type.)

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NOMENCLATURE, DESCRIPTION, & LOCATION	SEQ NO.	DESIGNATOR NO.	NEXT HIGHER ASSY	IDENTIFICATION NO. (FSC, FSN, P/N, ETC)	MANUFACTURER	SOURCE CODE STATUS	PAGE NO.
FAN, emergency exhaust (Rm 110)	MFV133-11	EEF-6			Trane		2T-42 thru 2T-46
MOTOR, emergency exhaust fan	MFV133-11A		MFV133-11				"
FAN, exhaust (Rm 106 roof)	MFV133-01	EF-1			"		2T-30 thru 2T-33
MOTOR, exhaust fan	MFV133-01A		MFV133-01		"		"
FAN, exhaust (Rm 110)	MFV133-02	EF-2			"		2T-42 thru 2T-46
MOTOR, exhaust fan	MFV133-02A		MFV133-02		"		"
FAN, exhaust	MFV133-03	EF-3			"		2T-40 thru 2T-41
MOTOR, exhaust fan	MFV133-03A		MFV133-03		"		"
FAN, exhaust (Rm 102)	MFV133-04	EF-4			"		2T-30 thru 2T-38
MOTOR, exhaust fan	MFV133-04A		MFV133-04		"		"

FIGURE 2. Index Page

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f. Use of commercial illustrations, photographs, and drawings to maximum possible extent.

g. Individual brochures that have a total thickness in excess of 2-1/2 inches shall not be divided to meet the maximum thickness criteria but shall be used intact and assigned a volume number as nearly as practicable in their applicable sequence.

3.1.9 Changes. Whenever changes are prepared, a "strip-in" change notice shall be added to the title page and the date of change inserted immediately below the basic date. See Figure 3 for example of a changed title page. Each new set of change pages shall include a sheet (format to be approved by the procuring agency) explaining in detail how to insert the change in the manual.

3.1.10 Drilling and Binding. Drilling shall be in accordance with MIL-P-38790. (Special drilling for three-post or three-ring binder.) Manuals shall be delivered unbound but contents shall be assembled and identified as specified herein.

3.2 Manner of Preparation. Manuals shall be prepared using an austere approach. Any extra effort deemed necessary by the contractor, such as preparation of supplemental data sheets, etc., shall have the prior approval of the procuring agency. (See 6.1)

3.3 Manufacturer's or Vendor Data. The manufacturer's or vendor's brochures and data shall be used whenever the technical content is adequate (see paragraph 3.4) or can be made adequate (see paragraph 3.5).

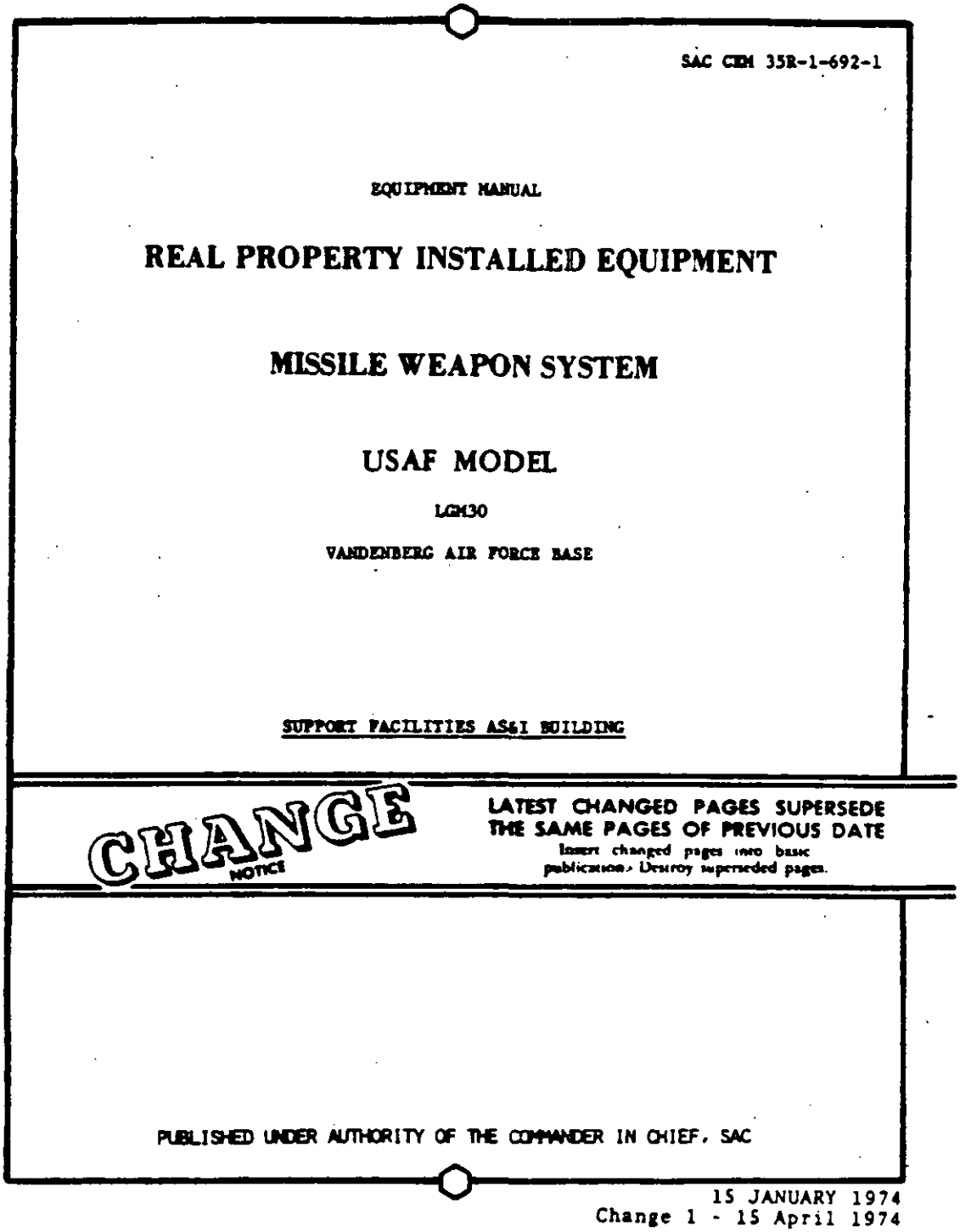
3.3.1 Depth of Coverage. Equipment manuals shall contain the applicable manufacturer's or vendor's equipment at the time of acceptance by the using command.

3.3.2 Duplicate Coverage. Duplicate coverage of manufacturer's or vendor's brochures/data shall be avoided where two or more identical pieces of equipment have been installed.

3.3.3 Arrangement of Information in Equipment Manuals. The information in the equipment manuals shall be separated into general RPIE system categories (air conditioning, heating, ventilating; power generation, and distribution, etc.). The data shall be arranged in alphabetical sequence according to the name of the manufacturer or vendor of the equipment within the system.

3.3.4 Technical Content. Each manufacturer's or vendor's brochure, bulletin, or other data shall be screened for adequacy and accuracy (see paragraph 3.4). If the manufacturer's data is found to be deficient, an attempt shall be made to obtain the required additional data from the manufacturer/vendor. If additional data is not available from the

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EQUIPMENT MANUAL

REAL PROPERTY INSTALLED EQUIPMENT

MISSILE WEAPON SYSTEM

USAF MODEL

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VANDENBERG AIR FORCE BASE

SUPPORT FACILITIES AS&I BUILDING

CHANGE NOTICE

LATEST CHANGED PAGES SUPERSEDE THE SAME PAGES OF PREVIOUS DATE

Insert changed pages into basic publication. Destroy superseded pages.

PUBLISHED UNDER AUTHORITY OF THE COMMANDER IN CHIEF, SAC

15 JANUARY 1974
Change 1 - 15 April 1974

FIGURE 3. Typical Title Page with Change Notice

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manufacturer/vendor, supplemental data shall be developed in accordance with paragraph 3.5.

a. Whenever feasible and economical, extraneous material shall be removed from brochures before inclusion in equipment manuals.

b. Operating, inspection, calibration, and cleaning instructions contained in manufacturer's data shall be changed if in conflict with developed procedures.

c. Each manufacturer's brochure, bulletin, or other data shall be checked to assure each item is covered by adequate data and configured as shown on the applicable shop drawing.

3.4 Equipment Operation and Maintenance Data. Manufacturer's operation and maintenance data is required for each end item of equipment and major components. Depth of this coverage shall be commensurate with the complexity of the item and probable level of maintenance which will be accomplished. Deviations in coverage and sequence are permitted when usability and maintenance support are not degraded. The following paragraphs define the requirements of this data.

3.4.1 Description of Equipment. The description shall include, where applicable, part number, model, size, weight, flow rate, amperage, etc.

a. An external view of the equipment, where not shown elsewhere, is necessary as well as text describing the function, method of operation, and principle features.

b. When the nomenclature or part number listing of the equipment manufacturer differs from that of the component manufacturer, both shall be shown.

3.4.2 Preparation for Use. Any special unpacking instructions, pre-installation lubrication, etc., shall be explained.

3.4.3 Operating Instructions.

3.4.3.1 Normal Operating Instructions. The instructions in this category shall be sufficient to enable the technician to adjust, stop and start, and operate the equipment properly. Special start-up precautions shall be noted, as well as other items requiring action before the equipment may be put into service. This information is normally covered in the manufacturer's installation manual.

3.4.3.2 Emergency Operating Procedures. The action to be taken in the event of a malfunction of the unit, either to permit a short period of continued operation, or to prevent further damage to the unit and to the system in which it is installed, shall be included.

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3.4.4 Maintenance Instructions. Maintenance instructions shall include the following as appropriate.

3.4.4.1 Cleaning and Lubrication. Instructions for periodic cleaning and lubrication, including type of lubricant and application points.

3.4.4.2 Troubleshooting. Instructions which state the indication of trouble, determination of cause, and instructions for the repair or replacement of worn or damaged parts necessary to restore the item to the required standards.

3.4.4.3 Inspection. Instructions for inspection of a disassembled item for damage or wear, tabular or chart form is preferred, wherever practicable, with emphasis upon allowable service limits, wear, backlash, end play, balance, length, depth of scoring, etc. These tolerances may exceed manufacturer's tolerances, but must not impair performance. This information shall be sufficiently comprehensive and adequate to serve as standards to determine when parts may be continued in use and when they must be replaced.

3.4.4.4 Corrosion Control. Instructions for peculiar corrosion prevention and/or removal when they are not covered by TO 1-1-2.

3.4.4.5 Disassembly and Reassembly. Instructions to provide the logical procedures necessary to disassemble and reassemble an item properly.

3.4.4.6 Repair. Repair procedures to bring the equipment up to a required standard, including instructions for examining equipment for needed repairs and adjustments, and any tests or inspections required to determine whether or not used parts may be reused or must be replaced.

3.4.4.7 Parts Lists. Lists that provide positive identification of parts necessary for the support of the equipment, to include sufficient information to enable personnel to requisition any needed replacement parts.

3.4.4.7.1 Listing. Part numbers and part names shown on illustrations or separately listed. When listed separately, they shall be keyed to applicable illustration by positively identifiable means.

3.4.5 Calibration Data for Test Equipment. Data which pertains to the use of special test equipment for maintenance, repair, calibration, or appraisal of operating conditions of a system, end item, or component shall contain specific instructions for calibration of the special equipment. These instructions shall include complete step-by-step procedures which enable the operator to check the reliability of the indications or readings provided by the test equipment.

a. Instructions shall outline precisely the steps required to restore the original accuracy of such indications or readings, and include the

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physical location of test connections and the values expected at these points. Whenever possible, instructions shall permit the accomplishment of calibration by utilizing a wiring harness, or exterior connections, or test points which are a part of, or auxiliary to, the special test equipment.

b. The calibration data furnished for special test equipment shall not depend upon the use of the end item that the equipment is to check, but shall, whenever possible, reflect simulated values wherein utilization of standard commercial equipment such as decode boxes, vacuum tube voltmeters, oscilloscopes, signal generators, weights, or measures can be used.

c. The data shall indicate specific tolerances, percentages of accuracy, range of scales, and method of use relative to the calibration equipment. Equipment referenced by make or model shall have prior approval of the procuring activity before being retained in the data.

3.5 Supplemental Data. When the procuring activity determines that existing commercial data will be satisfactory with the addition of supplemental data (see paragraph 3.2), the contractor shall prepare supplemental data to include the required information. These sheets shall be inserted in the manual immediately following the affected data, with a written or stamped note on the first page and at each applicable paragraph of the affected data directing the user to the supplemental data sheets, or may be incorporated by reworking, or interfiling with, the manufacturer's or vendor's data. The need for supplemental data will be determined on the basis of actual user need as related to complexity of hardware, etc. (See 6.1)

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for Compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the government to acceptance of defective material.

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4.2 Requirements. The quality assurance provisions shall be in accordance with MIL-M-38784 and MIL-P-38790.

5. PACKAGING

5.1 Packaging Requirements. Packaging, packing and marking for shipment shall be in accordance with MIL-P-38790 or as directed by the procuring activity.

6. NOTES

6.1 Data Requirements. When this specification is used in an acquisition which has a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved DD Form 1664, Data Item Description, and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.475-1 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements.

Para. No.	Data Requirement Title	DID No.	Option
3.1, 3.2, 3.5	Real Property Installed Equipment (RPIE) Manuals	DI-TMSS-80677	None

(Copies of data item descriptions required by the contractor in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting activity.)

6.2 Definitions.

6.2.1 Real Property. The fixed capital assets of the Government consisting of lands and interests herein, such as leases, permits, easements, licenses or rights-of-way, Government owned or leased grounds, structural facilities and facility plants and systems. It includes those items of nonexpendable equipment and property which are attached to or installed in real property.

6.2.2 Real Property Installed Equipment (RPIE). Items attached to or installed in real property. RPIE is normally programmed, procured, funded, and installed through the Military Construction Program (MCP).

6.2.3 Master Equipment List (MEL). A system oriented list of all RPIE end items of electrical and mechanical (includes hydraulic, pneudraulic, etc.) equipment and their major components.

6.2.4 MEL Sequence Number. The number assigned to each line item which indicates the sequence of the item in the MEL listing.

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6.2.5 Shop Drawings. Explanatory diagrams, brochures, catalog cuts, manufacturer's technical data and fabrication detail drawings provided for review by the contracting officer to determine conformance of the equipment or fabrication with the contract requirements.

6.3 Subject Term (Key Word) Listing.

Calibration Data
Equipment Manuals
Master Equipment List
Real Property Installed Equipment
System Manual

6.4 Changes from Previous Issue. Vertical lines or asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CUSTODIAN:

Air Force - 14

USER ACTIVITIES:

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PREPARING ACTIVITY:

Air Force - 14

(Project TMSS-F503)

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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